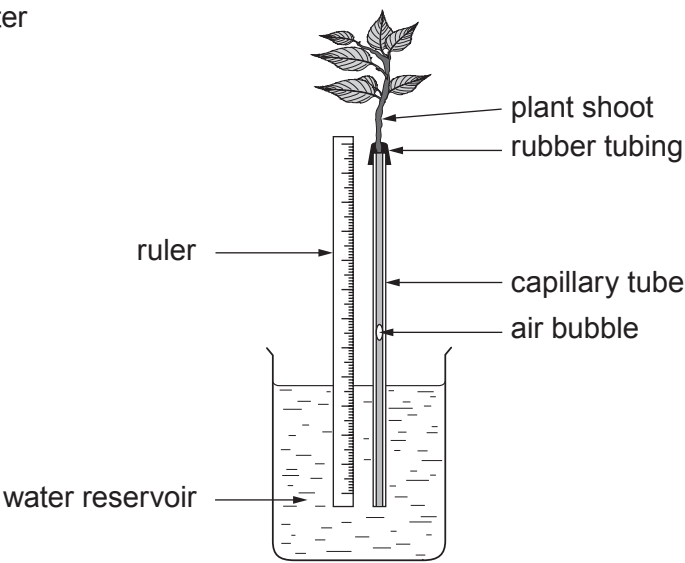


2. Students used a simple potometer as shown below to compare the rate of transpiration in shoots from a beech tree (*Fagus sylvatica*) and a bay tree (*Laurus nobilis*).

A simple potometer



(a) State the name of the tissue that transports water in plants. [1]

.....

(b) State **two** uses of water in plants. [2]

.....
.....

(c) (i) When the beech tree shoot was used, the air bubble in the capillary tube took 87 seconds to travel 50 mm. Calculate the rate of transpiration for the shoot from the beech tree in mm/s. [1]

Rate of transpiration = mm/s

(ii) When the bay tree shoot was used, the air bubble in the capillary tube took 51 minutes and 58 seconds to travel 50 mm.

State **one** conclusion that could be reached from the results. [1]

.....
.....



Examiner
only

(d) (i) State **one** factor that should have been kept constant in this investigation. [1]

.....

(ii) Describe how the students could improve their confidence in the results. [1]

.....

.....

(e) Explain why a potometer cannot be used to study transport of sugars in a plant. [2]

.....

.....

.....

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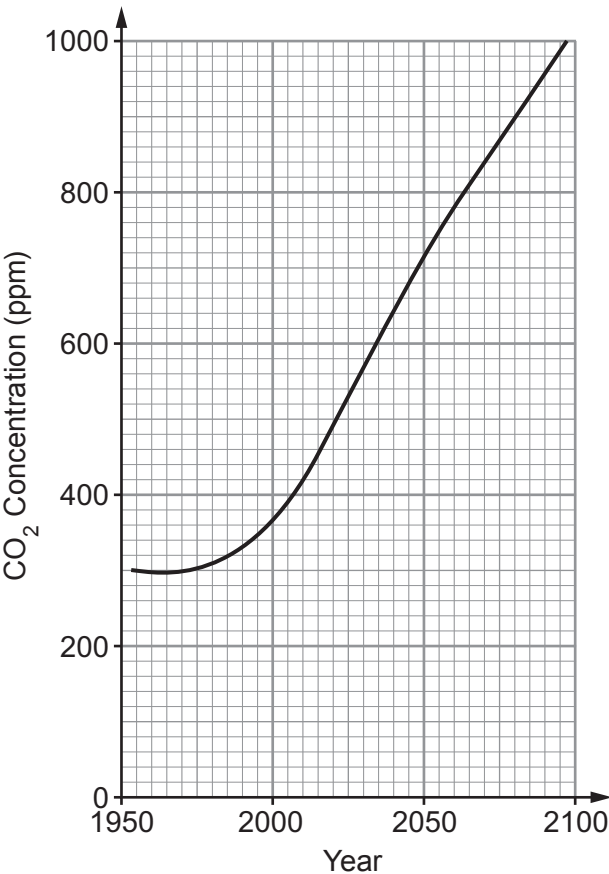
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9



The International Panel on Climate Change (IPCC) uses many different computer models to predict how CO₂ concentration in the atmosphere will change over time. The results of one of these models is shown in **Graph 2**.

Graph 2



Examiner
only

(a) (i) Write the word equation for photosynthesis. [2]

.....
(ii) State the name of **one** molecule made in plants from glucose. [1]

.....

(b) (i) Using both **Graphs 1** and **2**, estimate the predicted global uptake of carbon in 2080. [1]

.....
(ii) Suggest an explanation for the shape of **Graph 1** between 400 and 800 ppm. [1]

.....
.....

(c) (i) State how scientists increased the validity of their conclusions about the global changes to CO₂ concentrations in the atmosphere. [1]

.....
.....

(ii) Suggest why it is important that scientists use FACE experiments on plant growth to study the effects of increasing CO₂ concentrations instead of similar experiments in greenhouses. [1]

.....
.....

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END OF PAPER

